Fault Code: 145

Coolant Temperature Sensor Circuit

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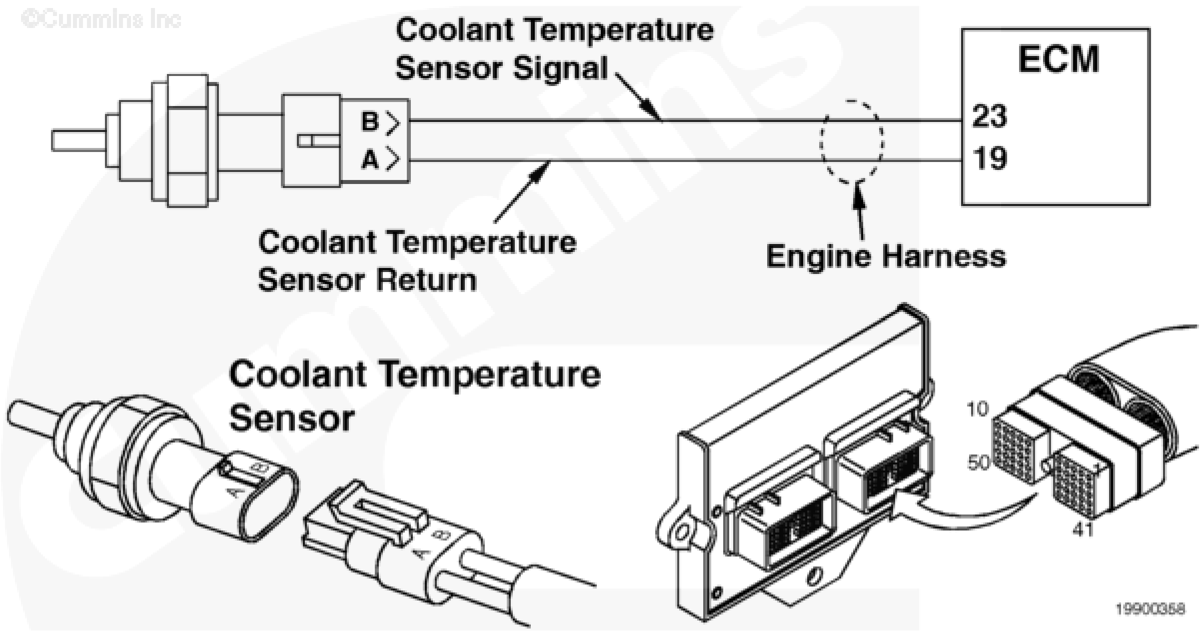
Overview

Codes	Reason	Effect
Fault Code: 145 PID(P): P110 SPN: 110 FMI: 4 Lamp: Yellow SRT:	Less than 0.21 VDC detected at the coolant temperature signal pin 23 of the engine harness.	Engine protection for coolant temperature is disabled.



Coolant Temperature Sensor Circuit

Circuit Description

The coolant temperature sensor is used by the electronic control module (ECM) to monitor the temperature of the engine coolant. The coolant temperature is used by the ECM for the engine protection system, timing, and fueling control.

The ECM monitors the voltage on pin 23. The ECM expects to see the voltage vary between 0.5 and 4.5 VDC. If the voltage is below 0.21 VDC for more than 2 seconds, then the ECM will log Fault Code 145.

Voltage below 0.21 VDC on pin 23 can be caused by short circuits to ground on the supply or return wires, or an internally grounded, failed sensor.

Component Location

The coolant temperature sensor is located on the left side of the engine in the thermostat housing.

Shoptalk

All temperature sensors:

- The resistance of the sensor varies with the temperature. The reading you observe will compare to the following table if the sensor is functioning properly.

Temperature (°C)	**Temperature [°F]**	**Resistance (ohms)**
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-145 (/qs3/pubsys2/xml/en/procedures/87/87-t05-145.html)

Last Modified: 29-Jul-2010